MCS-321	SOFTWARE PROJECT MANAGEMENT	L T P 3 1 0	4 Credits

Course Objectives:

CO1	To develop awareness regarding the theoretical and methodological issues related to software project management.
CO2	To develop software projects based on current technologies.
CO3	To make them understand the concepts of Project Management for planning to execution of projects.
CO4	To make them understand the feasibility analysis in Project Management and network analysis tools for cost and time estimation.
CO5	To enable them to comprehend the fundamentals of Contract Administration, Costing and Budgeting.

MODULE I

Overview of Project Management, PMI Processes, Software project phases, Organizational structures, Project charter, Statement of Work (SOW)
Planning Phase, Development lifecycle models, Matching lifecycles to projects, Project plans, Work Breakdown Structures (WBS)
Estimation and Budgeting, Estimation, Budgeting, Project, selection, NPV, ROI, Payback models

MODULE II

Scheduling , Project network diagram fundamentals, PERT techniques, Gantt charts, Critical chain scheduling
Risk and Change Management, Mid-term review , Risk management, Change control, More MS-Project
Development Management, Team models, Requirements process, Configuration , management, Software metrics, Programming languages & tools, Managing conflict and motivating, MS-Project: Assigning Resources.

MODULE III

Project Control, Status reporting, Project metrics, Earned value analysis, Communications Techniques, Process Improvement, MS Project: (a) Resource leveling (b) Other views
System Test Process, Test specifications, Black box and white box testing, Test scripts, Unit and integration testing, Acceptance test specifications, Test tools, MS Project:(a) Reporting
Final Phases & Other Issues, Project Recovery, Documentation, Cutover/Migration Post Project Reviews, Closing, MS Project: (a) Advanced features.

References

1. S. McConnell, "Software Project Survival Guide" (1997)

2. S. Berkun, "The Art of Project Management", (2005)

3. C. Larman, "Agile and Iterative Development: A Manager's Guide", (2003)

4. W. Royce, "Software Project Management: A Unified Framework", (1998)

5. J. Highsmith, "Agile Project Management: Creating Innovative Products", (2004)

6. T. DeMarco, "The Deadline: A Novel About Project Management", (1997)

7. T. DeMarco, "Peopleware: Productive Projects and Teams", (1999)

8. E. Bennatan, "On Time Within Budget: Software Project Management Practices and Techniques", (2000)

Course Outcome:

CO1	Understand the process of Software Project Management.
CO2	Conduct project planning activities that accurately forecast project costs.
CO3	Handle tools like MS Project & SVN
CO4	Understand the skills required for managing projects, project teams, and stakeholders
CO5	Understand the process of Software Project Management.